Keeping the Dog, Muting the Human: Speech Removal for Dog–Human Play Recordings

Nurmuhammed Kanybekov Péter Kovács
Faculty of Informatics, Eötvös Loránd University (ELTE), Budapest, Hungary
[draft of 22 October 2026 – not for distribution; author list and order to be confirmed]

Abstract

Recordings of dogs playing with people contain the people's speech, which must be removed before the recordings can be shared, while the dog vocalizations are the data. We study detect-and-mute anonymization on 125 minutes of annotated dog–human social play recorded at the ELTE BARKS Lab (30 sessions, 17 dogs, 581 human sound events, 2,055 dog vocalizations). Privacy is measured by speech recognizers acting as attackers, from two model families (Whisper small and large-v3-turbo, and NVIDIA Parakeet), and the cost by the dog vocalizations that are muted, overall and per call type. Every model is evaluated on dogs it never saw. A frame classifier on the frozen Whisper encoder combined with AudioSet class scores, trained to separate human sounds from dog vocalizations rather than from everything else, finds 84% of the human sound frames at a cost of 6% of the dog frames, against 72% for the previous AudioSet-only detector (74% for the best AudioSet-only variant). End to end, Whisper small reads two or more words in 1.0% of the human events afterwards (unprocessed: 19.8%; background noise alone: 1.0%), while 8.0% of the dog-only vocalization time is muted: a third less dog sound than with the previous detector (95% CI of the reduction: 21–51%), with no significant change in privacy for any of the three attackers. Speech enhancement instead of muting, three-microphone fusion and 20 ms decisions did not help. On synthetic scenes with unseen speakers, dogs and rooms, it leaves no utterance readable at the standard setting, as muting the exact utterances does, at a higher dog cost (57% vs 47% of the dog-only time). We release the code and a one-command anonymizer.

1 Introduction

Bioacoustic studies of dogs increasingly rely on recordings made during natural interaction with people. In dog–human social play the human partner talks, laughs and calls the dog throughout, so a recording that captures the dog's play pants, growls and barks also captures personal data: what was said, and a voice that identifies the speaker. Before such a dataset can be shared, the speech has to be removed. The dog sounds, however, are the point of the recording, and every second muted by mistake is data lost.

Removing speech from environmental recordings has so far been treated as a voice activity detection problem [1, 2]: detect speech, then delete or mute it. Speaker anonymization in the sense of the VoicePrivacy challenge [3] is a different task: it keeps the words and hides the identity. Here the words themselves must go. Two features of dog–human play make detection hard. First, the dog vocalizations are acoustically close to human non-speech sounds: a play pant is a breathy, forced exhalation [4], not unlike a human laugh or sigh. Second, the relevant cost is not generic false alarms but dog sound lost, so the detector should be judged at a fixed loss of dog vocalizations, not at a fixed false-positive rate on all audio.

This paper makes three contributions. (i) An evaluation protocol for this setting: leave-one-dog-out cross-validation, privacy measured by speech recognizers from two model families acting as attackers, and the cost measured as dog vocalization time and dog call events lost, per call type (Section 4). (ii) A detector that combines a frozen speech encoder (Whisper [5]) with AudioSet class scores [6] and is trained on the trade-off that matters, human sounds against dog vocalizations (Section 3). It mutes about as much speech as the previous AudioSet-only detector with a third less dog sound lost (Section 5). (iii) A set of negative results that narrow the design space: subtracting an estimate of the speech instead of muting it, fusing three microphones, and deciding every 20 ms instead of every 0.24 s all failed to improve the trade-off.

2 Data

The recordings were made at the ELTE BARKS Lab in Budapest between March and May 2025, during dog–human social play in a laboratory room: 30 sessions of 17 dogs with their human play partners, 124.7 minutes in total, recorded at 48 kHz with a shotgun (27 sessions) or cardioid microphone. Up to two further microphones recorded in parallel (Section 6). Expert annotators marked 581 human sound events (788 s; speech, laughter, claps and other human sounds) and 2,055 dog vocalizations of ten types. Panting dominates: 1,294 pants make up 3,521 s of the 3,966 s of annotated dog sound (89%). The other types are rarer: 198 growls, 163 grunts, 48 barks or yelps, 33 moans, 25 sneezes, 13 whines, 10 coughs or woofs, 7 shakes and 264 others.[check with the lab: dataset name and citation, annotation protocol, consent and permission to publish]

3 Method

Frames and features. The audio is resampled to 16 kHz and cut into 0.96 s windows every 0.24 s. Two frozen networks describe each window:

- *EfficientAT mn40* [6], a MobileNet distilled from transformer ensembles on AudioSet [7]. Its convolutional feature map is computed on 10 s chunks (the model saturates on 1 s inputs), averaged over the columns inside the window and passed through its classification head: 527 class scores.
- *Whisper small encoder* [5] (int8 ONNX export [8]). The audio is encoded in 30 s windows every 20 s, each 20 ms frame taken from the window in which it is most central (at least 5 s of context on both sides). The 768-dimensional frames inside the 0.96 s window are averaged. The export provides the first decoder layer's cross-attention keys, a fixed full-rank linear map of the encoder output, which carries the same information for a linear classifier. Whisper's representations are known to retain background-sound information [9]; here the question is whether they separate human sounds from dog sounds better than AudioSet scores.

Classifier and objective. A logistic regression ($C = 0.05$, balanced class weights, standardized inputs) is trained on frames that are at least half covered by a human event (positive) against frames with no human sound that are at least half covered by a dog vocalization (negative). Frames with neither are left out of training: muting background costs nothing the lab analyses, so the classifier should not spend capacity on it. The output is smoothed over 5 frames (± 0.48 s).

Thresholds and muting. Thresholds are set where a fixed share q of the dog-only frames is flagged (3, 6, 10 or 15%; the levels *light*, *standard* and *strict* use 3, 6 and 10%). Flagged frames are padded by 0.12 s and merged, and these intervals are set to zero in the original file. Everything else stays bit-identical.

4 Evaluation

Protocol. All results are leave-one-dog-out: each of the 17 dogs (with its human partner) is processed by a model trained on the other 16. Thresholds are set on the pooled out-of-fold predictions; for the AudioSet-only detector, choosing them inside the cross-validation instead gave the same results.

Privacy. A speech recognizer plays the attacker on each human event window (the event ± 0.3 s, loudness-normalized), before and after anonymization. We report the share of events where it reads two or more words, and the share of the words it read in the original that it can still read after muting (*words read back*). Sound tags such as "(dog barking)" are not counted. Three attackers are used: Whisper small (244M parameters) [5], Whisper large-v3-turbo (809M), and NVIDIA Parakeet TDT 0.6B v3 [10], a Fast-Conformer transducer trained on different data that covers 25 European languages including Hungarian. Parakeet matters because the detector itself uses Whisper's encoder. As a floor, the attackers also read background windows of the same length with no human sound: Whisper small writes 2+ words in 1.0% of them, Parakeet in 4.9% and turbo in 24.5%, so for turbo only *words read back* is meaningful. On background audio, the share of an event's original words that turn up is 0.0% for Whisper small and Parakeet and 1.6% for turbo.

Table 1: Frame level, leave-one-dog-out: share of human frames found when 3/6/10/15% of the dog-only frames are flagged, and ROC AUC (human vs dog-only).

input	trained vs	3%	6%	10%	15%	AUC
mn40	all	62.3	72.1	80.0	84.9	.927
mn40	dog	65.6	74.4	82.5	86.7	.938
Whisper	all	70.2	78.6	85.4	90.9	.948
Whisper	dog	72.9	80.7	87.5	92.2	.954
Whisper+mn40	all	71.2	81.2	86.6	90.8	.951
Whisper+mn40	dog	**75.9**	**84.0**	**89.4**	**93.2**	**.963**

Table 2: End to end, Whisper small as the attacker (581 events): events with 2+ readable words, original words read back, and dog-only vocalization time muted (all in %).

detector, dog frames flagged	readable	words	dog lost
none (unprocessed)	19.8	100	0
background only (floor)	1.0	0.0	–
VGGish [11], 6%	6.0	19.9	9.0
mn40, 6%	2.2	9.6	7.9
mn40, 10%	1.4	5.6	12.5
mn40, 15%	0.9	4.4	17.9
Whisper+mn40, 3% (light)	1.7	7.8	**4.3**
Whisper+mn40, 6% (standard)	**1.0**	5.2	8.0
Whisper+mn40, 10% (strict)	0.7	3.5	12.3
Whisper+mn40, 15%	0.5	2.6	17.8

Dog cost. The share of dog-only vocalization time (dog annotated, no human annotated) that is muted, and per call type the share of dog calls that remain *intact* (not a single sample muted). Calls that overlap a human sound are muted by design and reported separately.

5 Results

Frame level. Table 1 compares inputs and training objectives. Whisper's encoder alone beats the AudioSet scores at every dog cost, and the two together are best. Training against dog vocalizations only adds 1–5 points for every input. At 6% of the dog frames, the combined detector finds 84.0% of the human frames, against 72.1% for the previous detector (mn40, trained against all frames). At 3% dog cost it finds more human frames than the previous detector at 6%.

End to end. Table 2 shows the muted recordings as read by Whisper small. The new standard setting leaves 1.0% of the human events readable, the level Whisper reaches on background noise alone, and mutes 8.0% of the dog-only vocalization time. The previous detector needed 12.5% to reach 1.4%. A participant bootstrap (2,000 resamples of the 17 dogs) puts the reduction in dog sound lost at 36% (95% CI 21–51%); the confidence intervals of the privacy differences include zero (readable events: within ± 1.6 points; words read back: within ± 6 points). The same holds at the other two levels (light: -46%, CI 28–61%; strict: -31%, CI 16–45%).

Other attackers. Figure 1 repeats the comparison with all three recognizers. Parakeet, which shares nothing with

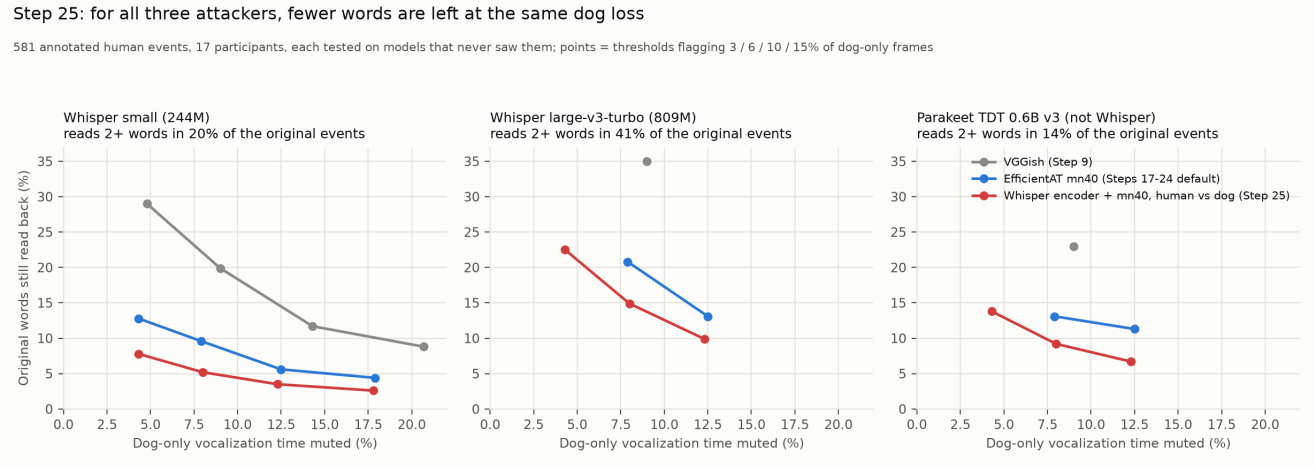


Figure 1: Original words read back after muting, for three attackers, against the dog-only vocalization time muted. Each line is one detector at 3/6/10/15% of the dog-only frames flagged (turbo and Parakeet were run at fewer points). On background audio, the share of an event's words that turn up is 0.0% (Whisper small, Parakeet) and 1.6% (turbo).

Table 3: Dog calls with no human sound on them that stay fully intact, previous vs new standard setting (%). Leave-one-dog-out.

call type	events	mn40, 10%	Whisper+mn40, 6%
pant	1,116	71	**82**
other	243	75	**83**
growl	188	90	**96**
grunt	159	66	**84**
bark/yelp	48	**75**	67
moan	32	62	**75**
whine	7	14	**29**

the detector, reads back 9.2% of its original words after the new standard setting and 11.3% after the previous one, while the dog loss drops from 12.5% to 8.0%. It reads 2+ words in 2.8% and 2.9% of the events, below its own background floor of 4.9% (10 events readable only after the previous setting, 9 only after the new one). Whisper large-v3-turbo reads back 14.9% of its words after the new standard setting and 13.1% after the previous one; at equal dog loss (new strict, 12.3%) it reads back 9.9%. For all three attackers the new detector's curve lies below the previous one: at the same dog loss fewer words are left. On the 29 events where Parakeet and Whisper small agree on at least two words of the original (clearly intelligible speech), one event remains readable after the new standard setting, against three after the previous one. The saving in dog sound therefore comes without a measurable loss of privacy for an attacker that shares nothing with the detector.

Dog calls by type. Because 89% of the dog time is panting, the time-based cost is mostly a panting number. Table 3 counts calls instead. The new detector leaves more calls intact for most types: 82% of the play pants (previously 71%), 84% of the grunts (66%). Barks are the main exception (67% vs 75%, 48 events): the speech encoder appears to respond to the harmonic, voice-like bark; sneezes also lose (83% vs 92%, 24 events). Whines (7 events) are mostly taken for a human voice by both detectors.

Whole recordings. The privacy numbers cover the annotated human events. To check for speech the annotators did not mark, Whisper read all 8 s windows with no annotated human sound within 0.5 s (519 windows, 68 minutes). Ten windows produced two or more words in the unprocessed audio: eight were typical hallucinations over dog sounds and two were short phrases worth a listen. No substantial unannotated speech was found.

Out of domain. The detector was also run, unchanged and with the thresholds set on ELTE, on 40 synthetic 30 s scenes built from material it never saw: read English speech from 20 LibriSpeech speakers [12], 20 ESC-50 dog clips [13], rooms simulated with pyroomacoustics [14] (RT60 0.2–0.8 s) and pink noise, with three utterances and four dog clips per scene that partly overlap. Whisper small, scored against the reference transcripts, reads two or more correct words in 99% of the unprocessed utterances. After the AudioSet-only detector it still does in 11.7% (light) and 7.5% (standard); after the new detector in 1.7% and 0%, as after muting the exact utterances (0%). Silero VAD [15] leaves 11.7%. The price is dog sound: the new standard setting mutes 57% of the dog-only time in these scenes, against 47% for the AudioSet detector and 47% for the exact utterances, because foreign barks overlap the speech and are hard to tell from it. An earlier attempt to reach this privacy by training the AudioSet detector on synthetic speech-under-bark scenes needed 63–77% dog loss.

6 What did not help

- *Subtracting instead of muting.* Where a dog overlaps the speech, subtracting a speech estimate (DTLN [16], or GTCRN [17] fine-tuned on this data) instead of muting keeps some dog sound, but at equal dog loss it never left fewer readable words than muting.
- *Three microphones.* Pooling the training data of all microphones and taking the maximum probability over them did not beat a single microphone with the same features.
- *20 ms decisions.* About 40% of the dog sound lost lies within 1 s of a human sound, which suggested finer boundaries. A classifier on every 20 ms Whisper frame (with ±0.06/0.5/1.5 s context, trained against all other

frames) muted 68% of the human time at 3.6% dog loss, against 73% at 4.5% for the 0.24 s detector with the same input and objective (77% at 7.1% vs 81% at 7.9%); interpolating the 0.24 s probabilities to 20 ms with tighter padding changed nothing.

- *Other inputs and labels.* Training on speech-only labels (events where Whisper reads words) instead of all human sounds, normalizing the scores per recording, adding neighbouring frames or the standard deviation of the Whisper frames were all equal or worse. Subtracting a separate dog detector's score helped only with a tuned weight, and less than changing the training objective.

7 Discussion and limitations

The largest gains came from the input (a speech encoder) and from the training objective (human vs dog), not from the classifier or the time resolution. The detector is a linear model on frozen features, trained on 16 dogs in a few minutes on a laptop, and the full anonymizer, including both frozen networks, runs at about 8× real time on two CPU cores.

The evaluation has limits. All real data come from one lab, one room and one kind of interaction; the synthetic out-of-domain test covers other speakers, dogs and rooms only in simulation. Privacy is measured by automatic recognizers on the annotated events; a human listening test of the remaining readable events is still to be done. The threshold fixes the dog cost on average: at the 6% setting the share of dog-only frames flagged ranges from 0% to 19% between dogs. Handler identity is not in the metadata, so if one person played with several dogs, the split is disjoint in dogs but not in speakers. The inputs, objective and settings were chosen over several rounds on the same 17 folds, which makes the best configuration somewhat optimistic; the out-of-domain test is the guard against this. Barks are muted more often by the new detector, which matters for studies that focus on barking; the AudioSet-only detector remains available for them. Finally, muting removes the dog sounds that overlap speech (by design); keeping those would require separating the two, which did not work well enough here.

8 Conclusion

For anonymizing dog–human play recordings, a frozen speech encoder combined with AudioSet class scores, trained on human versus dog sounds, reduces the readable speech to the level a recognizer reaches on background noise, while losing a third less dog vocalization than an AudioSet-only detector. The code, the evaluation and a one-command anonymizer that keeps the original file format are available at [repository URL once public].

Acknowledgements

[The ELTE BARKS Lab for the recordings and the annotations (annotators and reviewer as listed in the dataset metadata).]

References

[1] B. Cretois, C. M. Rosten, and S. S. Sethi, "Voice activity detection in eco-acoustic data enables privacy protection and is a proxy for human disturbance," *Methods in Ecology and Evolution*, 2022.

[2] D. Priebe, B. Ghani, and D. Stowell, "Efficient speech detection in environmental audio using acoustic recognition and knowledge distillation," *Sensors*, vol. 24, no. 7, p. 2046, 2024.

[3] N. Tomashenko *et al.*, "Introducing the VoicePrivacy initiative," in *Proc. Interspeech*, 2020.

[4] P. Simonet, D. Versteeg, and D. Storie, "Dog laughter: Recorded playback reduces stress related aggression in shelter dogs," in *Proc. 7th International Conference on Environmental Enrichment*, pp. 170–176, 2005.

[5] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. International Conference on Machine Learning (ICML)*, 2023.

[6] F. Schmid, K. Koutini, and G. Widmer, "Efficient large-scale audio tagging via transformer-to-CNN knowledge distillation," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

[7] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio set: An ontology and human-labeled dataset for audio events," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017.

[8] k2-fsa, "sherpa-onnx: speech recognition and related tools with ONNX runtime." `https://github.com/k2-fsa/sherpa-onnx`.

[9] Y. Gong, S. Khurana, L. Karlinsky, and J. Glass, "Whisper-AT: Noise-robust automatic speech recognizers are also strong general audio event taggers," in *Proc. Interspeech*, 2023.

[10] NVIDIA, "Now we're talking: NVIDIA releases open dataset, models for multilingual speech AI (granary, parakeet-tdt-0.6b-v3)." `https://blogs.nvidia.com/blog/speech-ai-dataset-models`, 2025.

[11] S. Hershey *et al.*, "CNN architectures for large-scale audio classification," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017.

[12] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: an ASR corpus based on public domain audio books," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2015.

[13] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. ACM International Conference on Multimedia*, 2015.

[14] R. Scheibler, E. Bezzam, and I. Dokmanić, "Pyroomacoustics: A python package for audio room simulation and array processing algorithms," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2018.

[15] Silero Team, "Silero VAD: pre-trained enterprise-grade voice activity detector." `https://github.com/snakers4/silero-vad`.

[16] N. L. Westhausen and B. T. Meyer, "Dual-signal transformation LSTM network for real-time noise suppression," in *Proc. Interspeech*, 2020.

[17] X. Rong, T. Sun, X. Zhang, Y. Hu, C. Zhu, and J. Lu, "GTCRN: A speech enhancement model requiring ultralow computational resources," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.